

**Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)**

**RU-100-01-00012
Object Oriented Programming with Java
SESSION:2025-2026**



**Submitted By :
RAVI KUMAR
ERP No.: RU-25-11681**

**Bachelor of Technology (Computer Science & Engineering
with AI and ML in Association with Google)**

**Under the Guidance of:
Mr. Ashish Shivhare**

**IN
RUNGTA INTERNATIONAL SKILLS UNIVERSITY
BHILAI, C.G.**

Project Title: E-Commerce Shopping Cart System

Abstract

The E-Commerce Shopping Cart System is a Java-based application that simulates the functionality of an online shopping platform. It allows users to add, remove, and view products in a cart while dynamically calculating the total price. The system applies discounts and taxes to generate a final bill.

This project is designed to demonstrate Object-Oriented Programming (OOP) concepts such as encapsulation, classes, and methods. It simplifies shopping operations and reduces manual calculation errors. The system can be further extended into a full web or GUI-based application.

Introduction

Online shopping systems require efficient cart management to handle multiple products and pricing logic. This project provides a simplified version of such a system using Java.

Features

- Add product to cart
 - Remove product
 - View cart
 - Auto price calculation
 - Apply discount and tax
 - Generate final bill
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Objectives

- Implement OOP concepts
 - Simulate real-world shopping logic
 - Perform dynamic calculations
 - Build structured Java classes
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Scope

This project is suitable for small-scale applications and learning purposes. It can be extended using:

- Database (MySQL)
 - GUI (Java Swing)
 - Web integration
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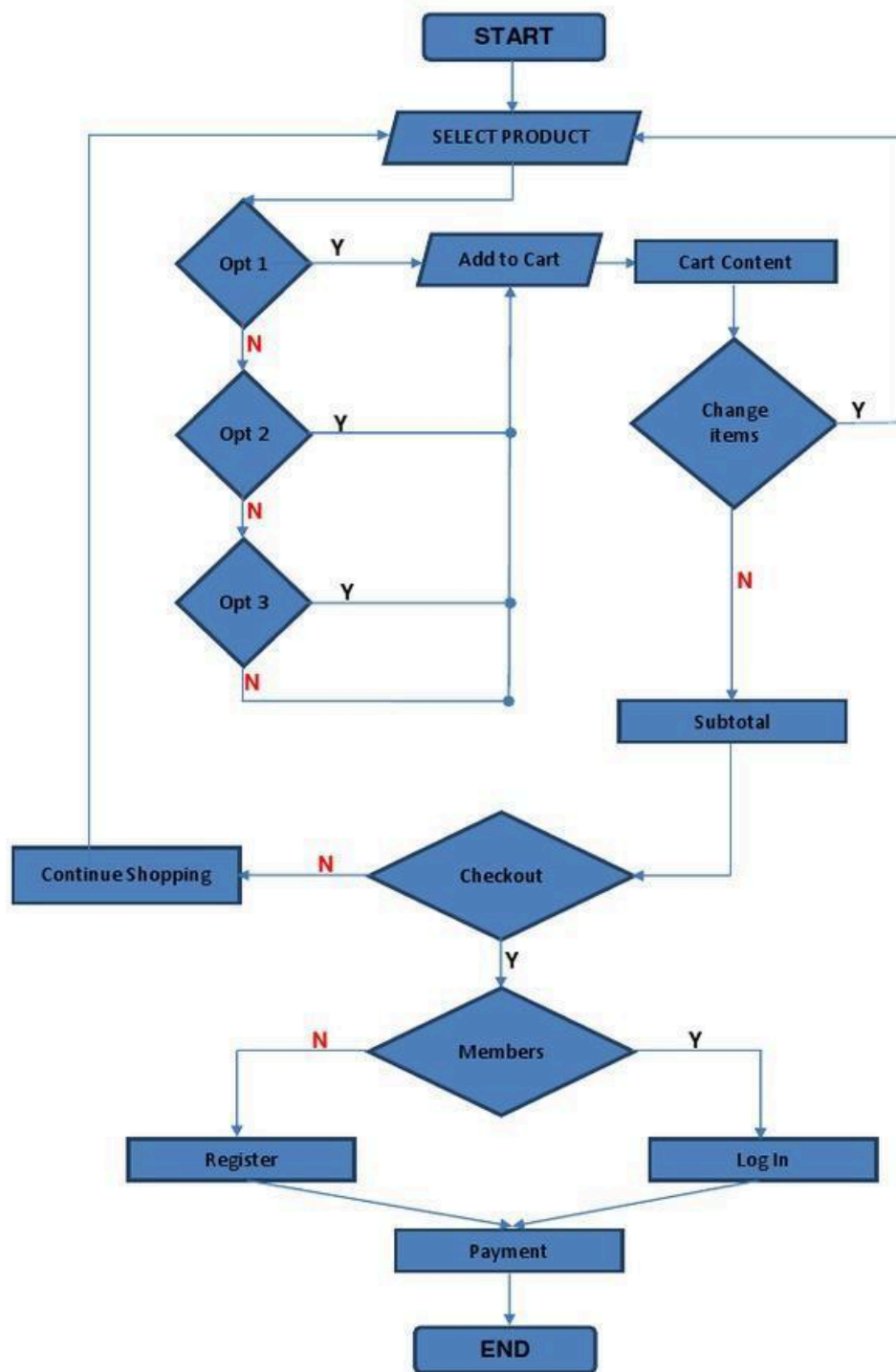
System Requirements

- **Hardware:** 4GB RAM minimum
 - **Software:** Java JDK, VS Code / Eclipse
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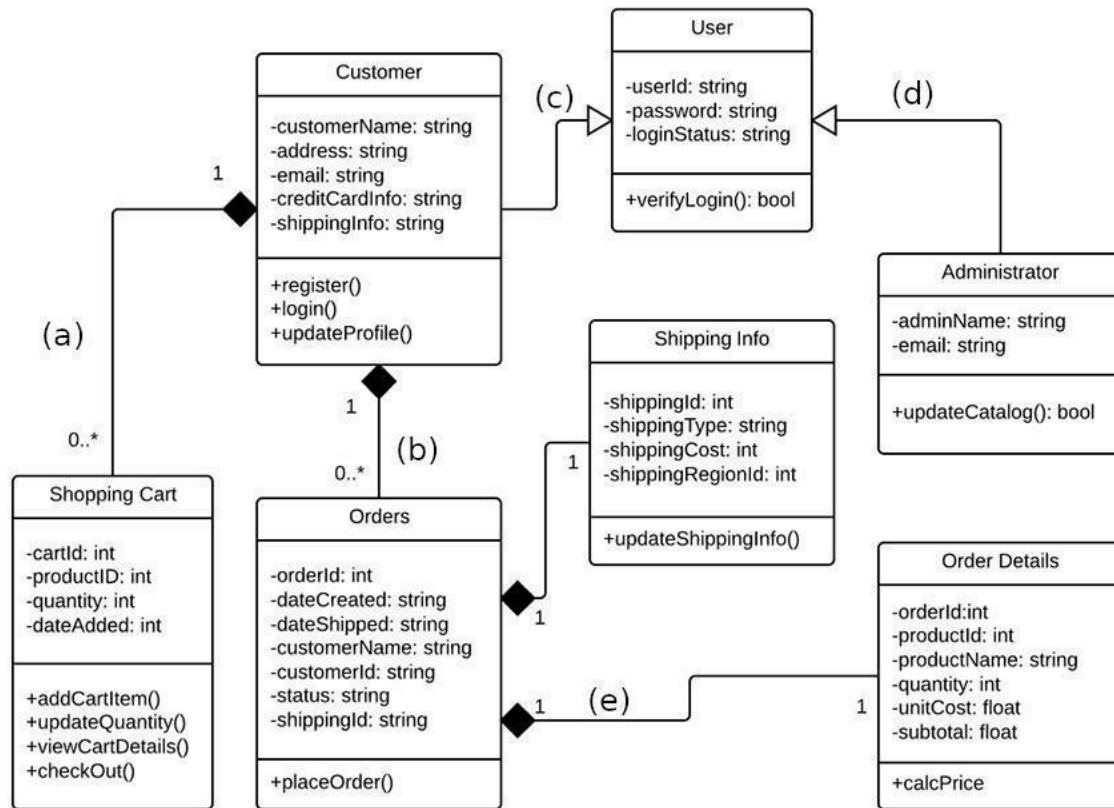
Methodology

1. User selects operation
 2. Add/remove product
 3. System updates total
 4. Apply discount & tax
 5. Display final bill
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Flowchart



Class Diagram



Implementation (Logic Explanation)

- **Product Class**: Stores product details
- **Cart Class**: Manages list and operations
- **Main Class**: Handles user input

Sample Input / Output

1. Shirt - ₹1000 × 2 = ₹2000
2. Shoes - ₹2000 × 1 = ₹2000

Total = ₹4000

Discount (10%) = ₹400

Tax (5%) = ₹180

Final Amount = ₹3780

Future Enhancements

- Login system
 - Coupon codes
 - GUI
 - Database storage
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Gantt Chart (Simple)

Task	Days
Planning	1
Design	2
Coding	3
Testing	2

Conclusion

The project successfully demonstrates OOP concepts and simulates a real-world shopping system with proper pricing logic.

References

- [1] H. Schildt, *Java: The Complete Reference*
- [2] Oracle Java Docs